LAYER 2 · DATA LINK

L2

Switching, VLANs and Loops

How switches move frames: VLANs and trunks, the MAC table, spanning tree, bundled links and locking ports down.

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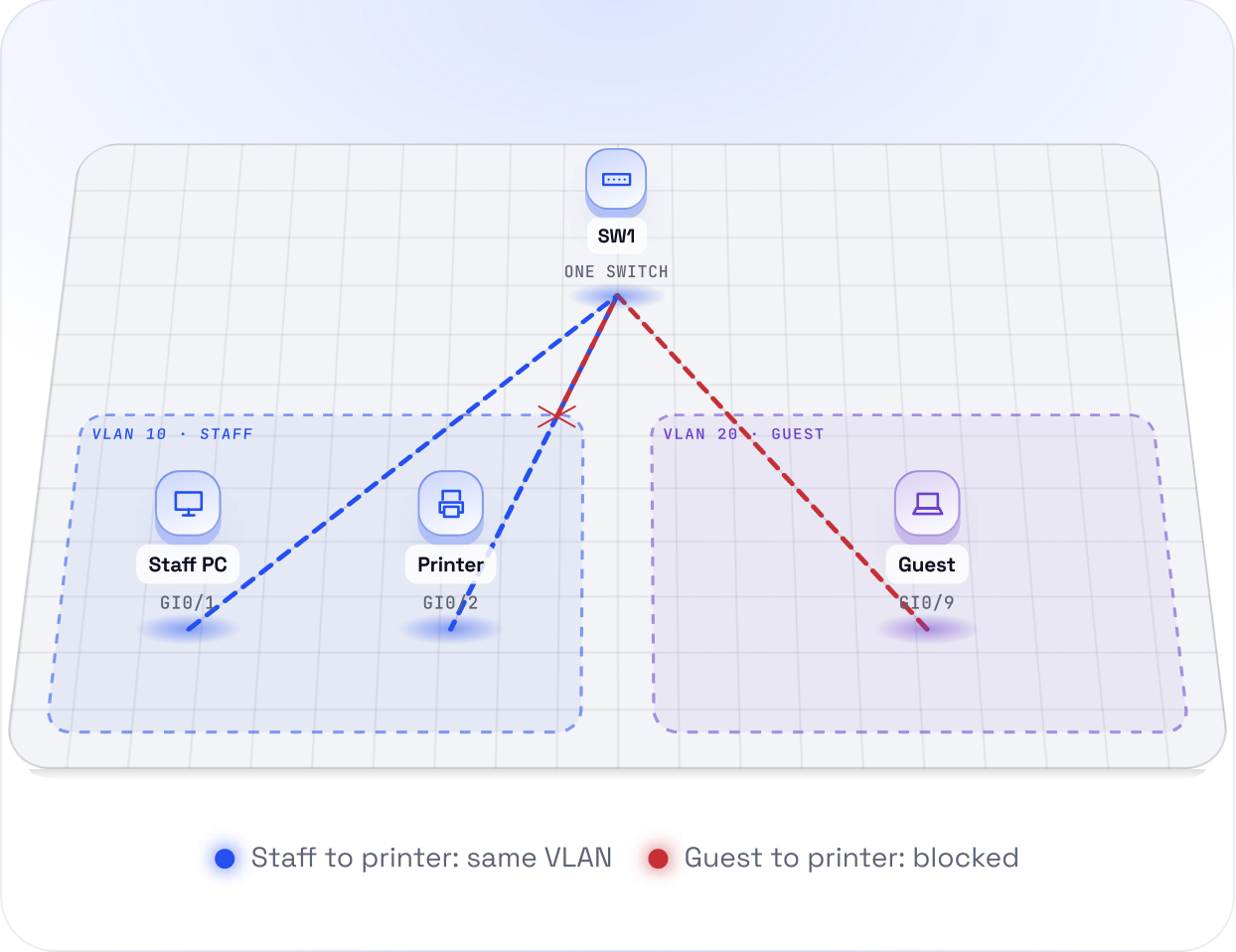
Limiting which devices may use a switch port, and what happens when another one plugs in.

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CHAPTER 1 · VLANS

VLANs

Splitting one physical switch into separate networks that cannot see each other.



Staff and guests share one switch but sit in different VLANs. Traffic between them never crosses without a router.

A VLAN (virtual LAN) splits one physical switch into several separate networks. Devices in different VLANs behave as if they were plugged into different switches: they cannot see each other's broadcasts, and they cannot talk at all without a router or a layer 3 switch in between.

Why bother

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on Wi-Fi should not be able to reach the accounts PC.

dcasts (ARP, DHCP) stay inside their own VLAN instead of reaching every device.

- **Order:** one switch can serve staff, guests, phones and cameras, each on its own network and subnet.

The usual rule is one VLAN, one subnet. VLAN 10 might be 10.10.10.0/24 and VLAN 20 10.20.20.0/24.

Configuring access ports

An **access port** belongs to exactly one VLAN and carries untagged frames for an ordinary device.

CISCO IOS

```
vlan 10
 name STAFF
vlan 20
 name GUEST
!
interface GigabitEthernet0/1
 switchport mode access
 switchport access vlan 10
!
interface GigabitEthernet0/9
 switchport mode access
 switchport access vlan 20
```

Check the result:

CISCO IOS

```
show vlan brief
show interfaces Gi0/1 switchport
```

Talking between VLANs

To let staff reach a shared printer in another VLAN, route between them: either a router with one sub-interface per VLAN ("router on a stick") or a layer 3 switch with an SVI per VLAN. Then use ACLs to decide which traffic is allowed across.

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> TIP

> Do not leave unused ports in VLAN 1. Put them in an unused "parking" VLAN and shut them down, so a stranger who plugs in gets nothing.

When it goes wrong

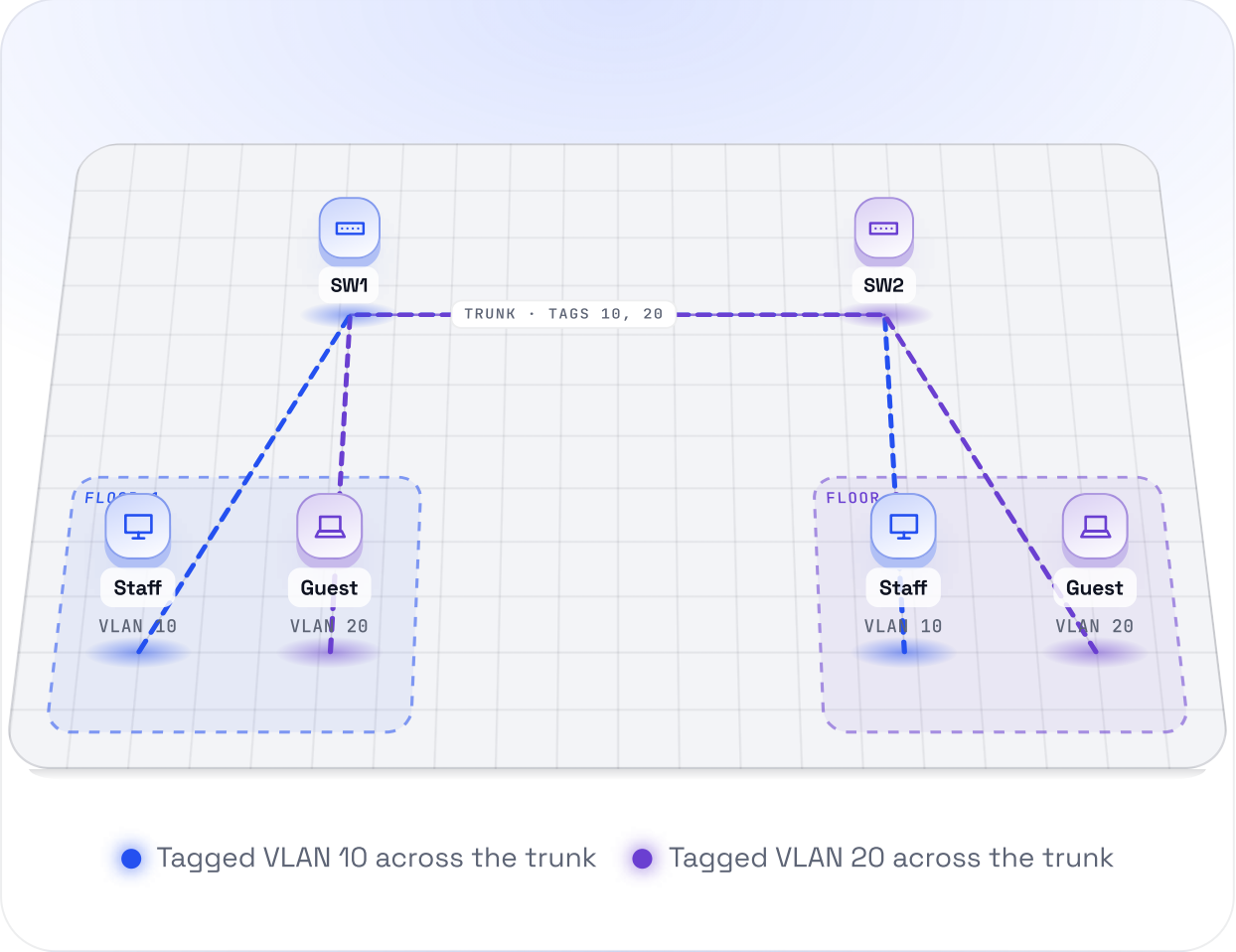
- **A PC gets a 169.254 address after moving desks:** its new port is in the wrong VLAN, so it never reaches the right DHCP scope.
- **Two PCs in the same VLAN cannot talk across switches:** the link between the switches is not a trunk, or the trunk does not allow that VLAN.
- **Everything works except one device:** compare `show interfaces switchport` for its port with a port that works.

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CHAPTER 2 · 802.1Q TRUNKING

802.1Q Trunking

Carrying many VLANs over one link by tagging each frame with its VLAN number.



One trunk carries both VLANs. Each frame is tagged on the way in and untagged at the far port.

When VLANs span more than one switch, the link between the switches has to carry all of them. Rather than run one cable per VLAN, a **trunk** carries every VLAN over a single link and marks each frame with the VLAN it belongs to. The standard for that mark is **IEEE 802.1Q**.

The tag

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byte tag into the Ethernet frame, between the source MAC herType. Inside it, 12 bits hold the **VLAN ID**, so a trunk can carry VLANs 1 to 4094.

1. A frame arrives untagged on an access port in VLAN 10.
2. The switch sends it out of the trunk **with a tag** saying VLAN 10.
3. The far switch reads the tag, removes it and delivers the frame only to ports in VLAN 10.

End devices never see the tag. Only switches, routers and servers configured for trunking do.

The native VLAN

One VLAN on a trunk travels **untagged**: the native VLAN, VLAN 1 by default. Both ends must agree on it. A mismatch leaks frames between VLANs and Cisco switches will log a CDP native VLAN mismatch.

Configuring a trunk

CISCO IOS

```
interface GigabitEthernet0/24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,99
```

The first line is only needed on switches that also support the old ISL standard. Then check:

CISCO IOS

```
show interfaces trunk
```

Look at three columns: the port is **trunking**, the native VLAN matches the other end, and your VLANs appear under "allowed and active".

WARNING

Set ports to `switchport mode access` or `switchport mode trunk` on [illegible]g them to negotiate with DTP means a device that [illegible] switch could form a trunk and see every VLAN.

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When it goes wrong

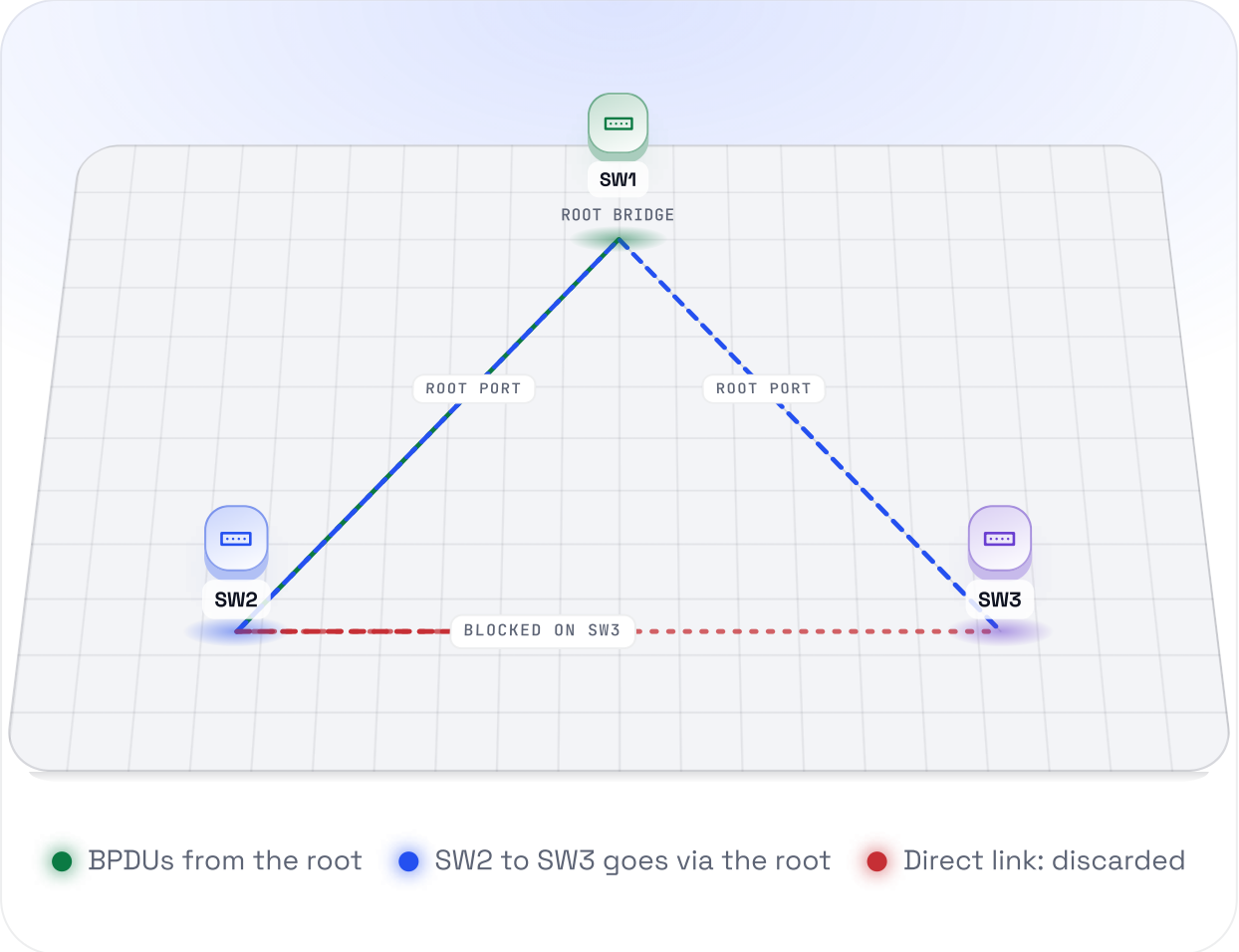
- **VLAN 20 works on one switch but not across the trunk:** VLAN 20 is missing from the allowed list, or was never created on the second switch.
- **Strange traffic in the wrong VLAN:** native VLAN mismatch between the two ends.
- **Trunk not forming:** one side is a static access port. Set both sides to trunk explicitly.

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CHAPTER 3 · STP

Spanning Tree (STP)

Keeping backup links without creating a loop that floods the network.



Three switches in a triangle. STP elects SW1 as root and blocks one port, so every frame has exactly one path.

Good networks have backup links. But Ethernet frames have no time-to-live, so if switches are joined in a loop, a single broadcast circles forever, multiplies and takes the whole network down within seconds. That is a **broadcast storm**. The Spanning Tree Protocol keeps the backup cables plugged in and blocks just enough ports to break every loop.

How STP decides

e small messages called **BPDUs**.

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1. **Elect a root bridge.** The switch with the lowest bridge ID wins. The bridge ID is a priority (32768 by default) followed by the MAC address, so on an untuned network the oldest switch often becomes root by accident.
2. **Pick a root port** on every other switch: the port with the lowest-cost path to the root.
3. **Pick a designated port** on every link: the side closer to the root forwards.
4. **Block everything else.** Those ports stay up but discard frames, ready to take over.

When a link fails, the blocked port moves to forwarding and traffic flows again.

Choose your root

Make your core or distribution switch the root on purpose:

CISCO IOS

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20 root primary
```

Rapid PVST+ (802.1w) converges in a second or two instead of the 30 to 50 seconds of classic STP, and runs one tree per VLAN.

Check the result:

CISCO IOS

```
show spanning-tree vlan 10
```

The output lists the root bridge, then each port with its role (Root, Desg, Altn) and state (FWD, BLK).

Edge ports

Ports that connect to PCs and printers never create loops. Mark them as edge ports so they start forwarding immediately, and protect them in case someone plugs a switch in:

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```
interface range GigabitEthernet0/1 - 20
 spanning-tree portfast
 spanning-tree bpduguard enable
```

> TIP
>
> "The PC takes 30 seconds to get an IP address after boot" is very often a port without PortFast. DHCP gives up before the port starts forwarding.

When it goes wrong

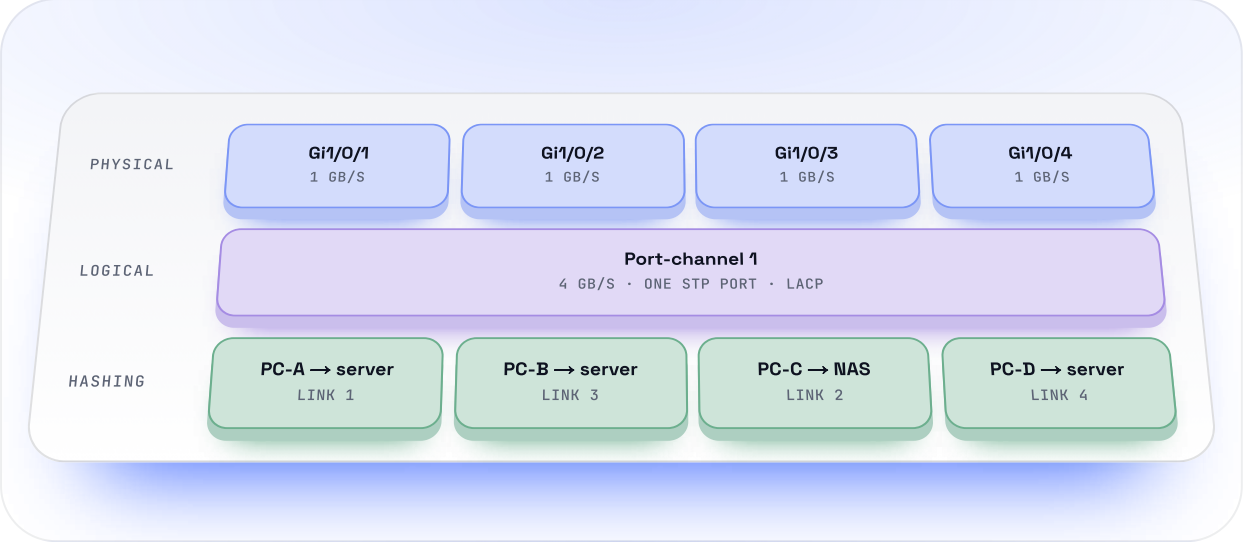
- **Network crawls, switch lights flash solid:** a loop with STP disabled or a hub in the path. Unplug links until it stops, then find the loop.
- **Traffic takes a strange path:** the wrong switch is root. Set the priority deliberately.
- **A port is err-disabled:** BPDU Guard did its job; someone connected a switch to a desk port.

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CHAPTER 4 · ETHERCHANNEL

EtherChannel

Bundling several links into one logical link for more bandwidth and no blocked ports.



Four physical ports become one Port-channel. Each conversation is hashed onto one member link.

Two switches joined by four cables would normally have three of them blocked by spanning tree. **EtherChannel** bundles those links into one logical link, a **Port-channel**, so all four carry traffic and STP sees a single port.

What you gain

- **Bandwidth:** four 1 Gb/s links give up to 4 Gb/s in total.
- **Resilience:** if one cable fails, traffic moves to the others in under a second, with no spanning tree recalculation.
- **Simplicity:** one interface to configure and monitor instead of four.

Load balancing is per conversation

EtherChannel does not split a single file transfer across all links. It **hashes** each [illegible] fields such as source and destination MAC or IP addresses, [illegible]versation on one member link. Many users spread nicely across the bundle; one big transfer still tops out at the speed of one link.

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CISCO IOS

```
port-channel load-balance src-dst-ip
```

Building it with LACP

LACP (802.3ad) is the open standard. Both ends negotiate, and a miscabled link is left out instead of causing a loop.

CISCO IOS

```
interface range GigabitEthernet1/0/1 - 4
 channel-group 1 mode active
!
interface Port-channel1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,99
```

Use `active` on at least one side (`active` or `passive` on the other). Then check:

CISCO IOS

```
show etherchannel summary
```

Look for `(SU)` on the Port-channel (layer 2, in use) and `(P)` on each member (bundled).

The rule that matters

Every member port must match: same speed, same duplex, same mode (access or trunk), same allowed VLANs and same native VLAN. Configure the Port-channel interface, not the members, and the settings stay consistent.

When it goes wrong

`(s)` or `(I)`: suspended or standalone, usually a mismatch or LACP mode on the far end.

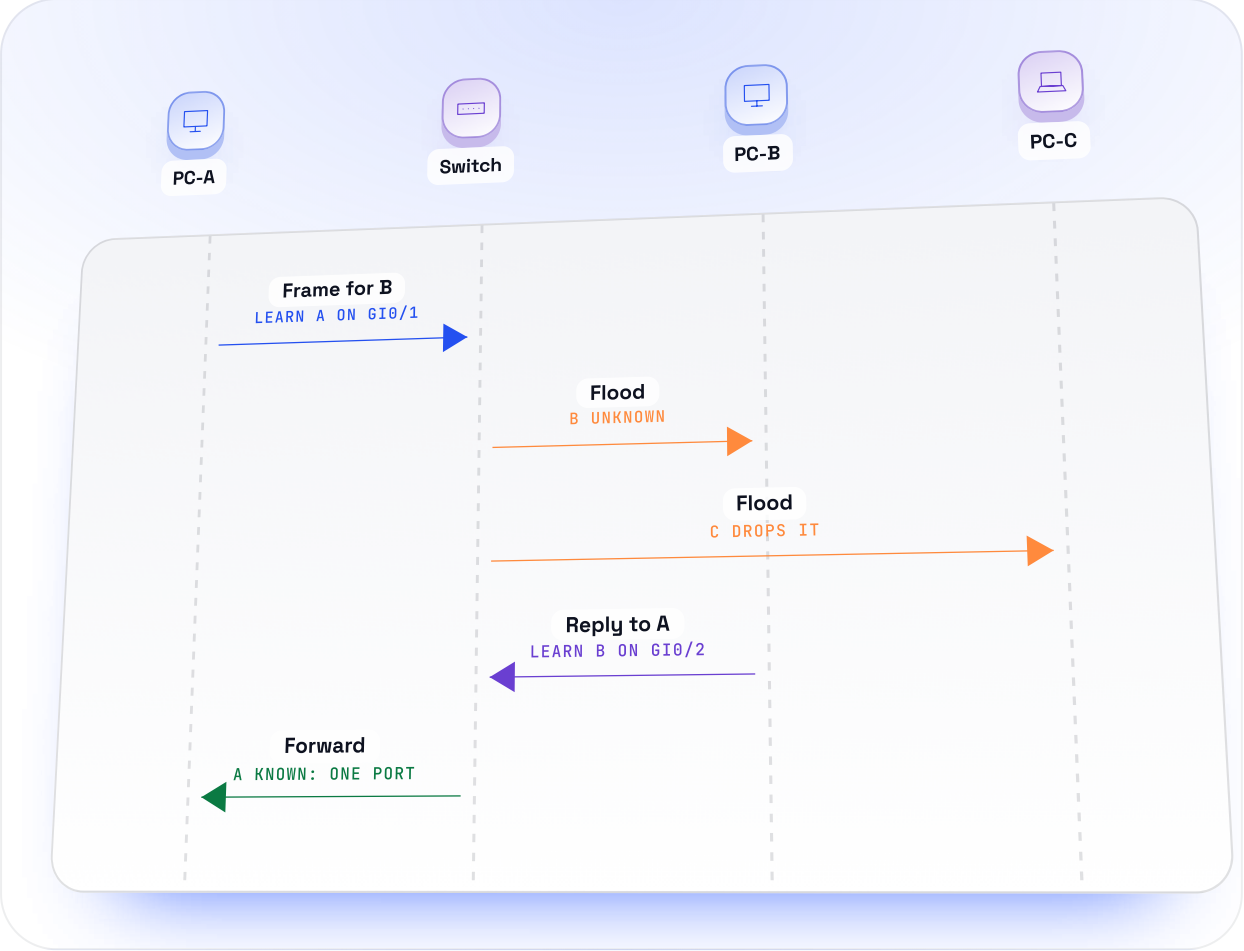
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- **Bundle up but slower than expected:** traffic is between few hosts, so the hash puts it on one link. Try a different load-balance method.
- **Loop or err-disable when cabling:** one side was configured with `mode on` and the other was not. Prefer LACP.

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MAC Address Tables

How a switch learns where each device is, and why it floods when it does not know.



The switch learns sources as frames arrive, floods frames for unknown destinations, then forwards directly.

A switch decides where to send each frame by looking up its destination MAC address in a table. Nobody types that table in. The switch builds it by watching traffic.

Learn, flood, forward, filter

1. **Learn:** when a frame arrives, the switch records its **source** MAC address it came in on.
2. **tination** MAC is not in the table yet (or is the broadcast address), the switch sends the frame out of every port in that VLAN except

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the one it came from.

3. **Forward:** if the destination is known, the frame goes out of that one port only.
4. **Filter:** if the destination is on the same port the frame came in on, the switch drops it.

After the first reply, both devices are in the table and their traffic goes straight between their two ports.

Reading the table

CISCO IOS

```
show mac address-table
show mac address-table address 0050.56ab.12cd
show mac address-table interface GigabitEthernet0/5
```

VLAN	MAC ADDRESS	TYPE	PORTS
10	0050.56ab.12cd	DYNAMIC	Gi0/5
10	00e0.4c68.0a11	DYNAMIC	Gi0/24
20	a4bb.6d10.77f2	STATIC	Gi0/9

A port with many MAC addresses behind it, like Gi0/24 often is, leads to another switch or an access point.

Ageing

Dynamic entries expire after **300 seconds** without traffic, so the table follows devices that move or switch off.

Finding a device on the network

This is one of the most useful support tricks. If you know a device's IP address:

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`arp -a` on a machine in the same subnet (or `show ip arp` on the gateway) to get its MAC address.

2. Run `show mac address-table address <mac>` on the core switch to see which port it is behind.
3. Follow that port to the next switch and repeat until you reach an access port with a single MAC. That is the desk.

When it goes wrong

- **A MAC address flaps between two ports:** a loop, or the same device connected twice.
- **The table fills up and everything floods:** a MAC flooding attack or a faulty device. Port security limits how many addresses a port may learn.

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CHAPTER 6 · PORT SECURITY

Port Security

Limiting which devices may use a switch port, and what happens when another one plugs in.



The desk PC's MAC is learned as sticky. When an unknown laptop appears on a secured port, the port shuts down.

Anyone can walk up to a desk, unplug the PC and plug in their own laptop. **Port security** lets a switch port accept only the devices it should, and react when something else appears.

What it controls

- **How many** MAC addresses may use the port (usually 1, or 2 for a phone with a
- s: typed in by hand, or learned automatically and remembered (**sticky**).

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- **What happens** on a violation.

Violation modes

MODE	UNKNOWN FRAMES	LOGS IT	PORT STAYS UP
`protect`	dropped	no	yes
`restrict`	dropped	yes	yes
`shutdown`	dropped	yes	no, err-disabled

`shutdown` is the default and the safest: the port goes dark until someone checks it.

A typical desk port

CISCO IOS

```
interface GigabitEthernet0/5
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
```

Port security only works on ports set statically to access or trunk mode, which is why the first line matters.

Check it:

CISCO IOS

```
show port-security interface Gi0/5
show port-security address
```

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rt back

When a port is err-disabled, find out why first. If it was a genuine new PC, clear the old sticky address and re-enable the port:

```
CISCO IOS
clear port-security sticky interface Gi0/5
interface GigabitEthernet0/5
 shutdown
 no shutdown
```

Or let ports recover on their own after a few minutes:

```
CISCO IOS
errdisable recovery cause psecure-violation
errdisable recovery interval 300
```

> **NOTE**
>
> Port security stops casual misuse, not a determined attacker, who can copy an allowed MAC address. For stronger control, use 802.1X, which checks a certificate or login before the port opens.

When it goes wrong

- **"My new laptop doesn't work at my desk":** the port still remembers the old PC's sticky MAC.
- **A phone and PC on one port, one of them blocked:** raise `maximum` to 2.
- **Ports keep shutting at night:** a docking station or VM is presenting extra MAC addresses.